

The EFGP M-step: the collapsed objective, and exactly what is dropped

Companion to `efgp_estep.tex` (same notation)

Scope. The E-step doc analyses the transition term of the ELBO (1) at fixed hypers. This note does the *M-step*: hold $q(x)$ (hence its sufficient statistics) fixed and maximise the ELBO over the kernel hypers $\theta = (\ell, \sigma_f^2)$. We (1) derive the collapsed objective (7) from scratch, (2) show it is the *exact* GP marginal likelihood of the pseudo-velocity regression — it drops nothing θ -dependent except a constant, and (3) locate precisely the terms EFGP *does* leave out: they live in the E-step that *builds* $q(x)$, not in this objective.

1 The ELBO and the M-step objective

Recall the structured posterior $q(x_{0:T})q(f)$, independent GP drifts $f_r \sim \mathcal{GP}(0, k_\theta)$, diagonal diffusion $\Sigma = \text{diag}(\sigma_1^2, \dots)$, Euler transition $x_{i+1} = x_i + \Delta_i f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \Delta_i \Sigma)$, and $\delta_i := x_{i+1} - x_i$. The ELBO is

$$\mathcal{L} = \mathbb{E}_{q(x)}[\log p(y | x)] + \sum_i \mathbb{E}_{q(x,f)}[\log p(x_{i+1} | x_i, f)] + \mathbb{E}_{q(f)}[\log p(f) - \log q(f)] + H[q(x)] + \mathbb{E}_{q(x_0)}[\log p(x_0)]. \quad (1)$$

Only the transition term and the $q(f)$ -KL depend on θ ; the emission, entropy and initial terms are θ -free. So the M-step objective is

$$\mathcal{L}_M(\theta) = \sum_i \mathbb{E}_{q(x,f)}[\log p(x_{i+1} | x_i, f)] + \mathbb{E}_{q(f)}[\log p(f) - \log q(f)] + (\theta\text{-free}). \quad (2)$$

Notation (the rest lives in `efgp_estep.tex`; D = latent dimension).

D, D_{out}	latent/input dimension; number of drift output coordinates ($D_{\text{out}}=D$ here).
m_i, S_i	mean, covariance of $q(x_i) = \mathcal{N}(m_i, S_i)$; $S_{i,i+1} = \text{Cov}(x_i, x_{i+1})$.
$d_i := m_{i+1} - m_i$	mean increment; $C_i := S_{i,i+1} - S_i$ (Stein cross-cov correction).
ξ_k, M	spectral-grid frequencies ($k=1:M$) and grid size.
F_x, D_θ	Fourier features $F_x[k] = e^{2\pi i x^\top \xi_k}$; $D_\theta = \text{diag}(\sqrt{S(\xi_k; \theta)})$ (real, diagonal).
$\phi_\theta(x) = D_\theta F_x$	drift basis; $f_r(x) = \phi_\theta(x)^* w_r$, whitened weights $w_r \sim \mathcal{CN}(0, I)$.
$\bar{f}_r(x) = \phi(x)^* \mu_r$	posterior-mean drift; $J_{\bar{f}}$ its Jacobian, $H_{\bar{f}_r}$ the Hessian of coord. r .
$V_r(x) = \sigma_r^{-2} \phi(x)^* A_r^{-1} \phi(x)$	$q(f_r)$ posterior-variance at x .
$A_r, h_r, \mu_r = A_r^{-1} h_r$	$q(f_r)$ precision, RHS, mean in the basis (6).
$\mathcal{CN}(0, \cdot)$	circularly-symmetric complex normal.
$\mathbb{E}_{q(x_i)}, \mathbb{E}_{q(x_i, x_{i+1})}$	expectation under $q(x_i)$, resp. the pair $q(x_i, x_{i+1})$.

2 Collapsing $q(f)$: derivation of the objective

Work one drift coordinate r at a time (they decouple; drop r). From `efgp_estep.tex` eq. (f-quadratic), the only f_r -dependent part of the transition log-density at transition i is $\sigma_r^{-2} \delta_{i,r} f_r(x_i) - \frac{\Delta_i}{2} \sigma_r^{-2} f_r(x_i)^2$. Put f_r in a finite basis with a *whitened* weight w_r (unit prior),

$$f_r(x) = \phi_\theta(x)^* w_r, \quad \phi_\theta(x) = D_\theta F_x, \quad F_x[k] = e^{2\pi i x^\top \xi_k}, \quad w_r \sim \mathcal{CN}(0, I), \quad (3)$$

where $D_\theta = \text{diag}(\sqrt{S(\xi_k; \theta)})$ carries the RBF spectral density

$$S(\xi; \theta) = \sigma_f^2 (2\pi \ell^2)^{D/2} e^{-2\pi^2 \ell^2 \|\xi\|^2} \quad (\text{efgp_jax_drift.py:ws_real_se}). \quad (4)$$

Taking the $q(x)$ -expectation of the likelihood (the expectation touches only the features, never w_r) gives a Gaussian in w_r :

$$\mathbb{E}_{q(x)}[\log p(\cdot \mid w_r)] = \text{Re}\langle h_r, w_r \rangle - \frac{1}{2} w_r^* (A_r - I) w_r - \frac{1}{2} \sum_i \frac{1}{\Delta_i} \mathbb{E}_{q(x_i, x_{i+1})} [\delta_{i,r}^2], \quad (5)$$

with (E-step doc eq. (Ah))

$$A_r = I + \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}_{q(x_i)} [\phi(x_i) \phi(x_i)^*], \quad h_r = \frac{1}{\sigma_r^2} \sum_i \mathbb{E}_{q(x_i, x_{i+1})} [\phi(x_i) \delta_{i,r}]. \quad (6)$$

Adding the unit prior $-\frac{1}{2} w_r^* w_r$ turns $A_r - I$ into A_r , and the KL term $\mathbb{E}_{q(f)}[\log p - \log q]$ is what the collapse of the Gaussian integral produces. Marginalising w_r (complex Gaussian, $\int e^{-\frac{1}{2} w^* A w + \text{Re } h^* w} dw \propto |A|^{-1} e^{\frac{1}{2} h^* A^{-1} h}$) and negating gives the per-dim loss; summing over the D_{out} coordinates,

$$\mathcal{L}_M(\theta) = \frac{D_{\text{out}}}{2} \log |A_\theta| - \frac{1}{2} \sum_r \text{Re}\langle h_{\theta,r}, \mu_{\theta,r} \rangle, \quad \mu_{\theta,r} = A_\theta^{-1} h_{\theta,r}, \quad (7)$$

which is exactly what `m_step_kernel_jax` minimises (`efgp_jax_drift.py:773--796`), using $\langle \mu, A\mu \rangle = \langle \mu, h \rangle$ to collapse the quadratic. This is the E-step doc’s eq. (Mstep-collapsed).

How θ enters. On the fixed Fourier grid the features F_x , and therefore the assembled statistics

$$T := \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}_{q(x_i)} [F_{x_i} F_{x_i}^*] \quad (\text{the } \theta\text{-free Gram, top}), \quad z_r := \frac{1}{\sigma_r^2} \sum_i \mathbb{E}_{q(x_i, x_{i+1})} [F_{x_i} \delta_{i,r}] \quad (\mathbf{z}, \mathbf{r}), \quad (8)$$

do *not* depend on θ . All θ -dependence is through the prior weights D_θ : $A_\theta = I + D_\theta T D_\theta$ and $h_{\theta,r} = D_\theta z_r$ (`efgp_jax_drift.py:777,786`). So (7) is precisely a GP *marginal likelihood* in which the “data” (T, z_r) — the frozen $q(x)$ sufficient statistics — are held fixed and only the kernel varies. That is the formal content of “the M-step is a GP hyperparameter update on the reconstructed path.”

3 Reading the objective as GP marginal likelihood

(7) is the standard $-\log$ evidence of a linear-Gaussian model:

$$\underbrace{-\frac{1}{2} \sum_r \text{Re}\langle h_r, A_\theta^{-1} h_r \rangle}_{\text{data fit}} + \underbrace{\frac{D_{\text{out}}}{2} \log |A_\theta|}_{\text{Occam / complexity}}. \quad (9)$$

The complexity term carries the collapsed $q(f)$ posterior variance exactly: with $V_r(x) = \sigma_r^{-2} \phi(x)^* A_r^{-1} \phi(x)$ the drift-posterior variance at x ,

$$\sum_i \Delta_i \mathbb{E}_{q(x_i)}[V_r(x_i)] = \sigma_r^{-2} \sum_i \Delta_i \operatorname{tr}(A_r^{-1} \mathbb{E}_{q(x_i)}[\phi \phi^*]) = \operatorname{tr}(A_r^{-1}(A_r - I)) = M - \operatorname{tr}(A_r^{-1}), \quad (10)$$

the same information (a monotone function of the eigenvalues of A_θ) that $\log |A_\theta| = \sum_k \log \lambda_k$ encodes (λ_k the eigenvalues of A_θ). **So the drift-posterior variance is present in the M-step objective**; it is not a dropped term here.

4 Exactly what is left out

Two categories. Be precise about which is which.

(A) Dropped from the objective (7) — one term, and it is harmless

The **red** term in (5),

$$\frac{1}{2} \sum_i \frac{1}{\Delta_i} \mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \delta_i] \quad (\text{piece (i) of `efgp_estep.tex` eq. (transition-frozen)}), \quad (11)$$

is the “ $y^\top y$ ” self-energy of the pseudo-velocity targets. It depends on $q(x)$ (the increments δ_i) but *not on* θ , so it is an additive constant in the M-step and `m_step_kernel_jax` correctly omits it. Dropping it does not move $\arg \max_\theta$. Nothing else θ -dependent is dropped from (7): the increment fit (via z_r), the Stein cross-covariance correction $C_i = S_{i,i+1} - S_i$ (folded into z_r , E-step doc eq. (hr-stein)), the input uncertainty S_i , and the $q(f)$ variance (10) are all retained.

Input uncertainty is retained, via the characteristic function. θ -frozen though T is, it is *not* the point Gram: each $\mathbb{E}_{q(x_i)}[\phi \phi^*]$ uses (E-step doc eq. (Eqx-chi))

$$\mathbb{E}_{q(x_i)}[e^{2\pi i x_i^\top \xi}] = e^{2\pi i m_i^\top \xi} \underbrace{e^{-2\pi^2 \xi^\top S_i \xi}}_{\text{Gaussian envelope}}, \quad (12)$$

so S_i enters T (and z_r) as a frequency-domain damping. At the small S_i of a well-identified latent ($\operatorname{tr} S_i \sim 5 \times 10^{-3}$ here) the envelope ≈ 1 at the informative frequencies $\|\xi\| \sim 1/\ell$, so the M-step is *numerically* close to the point ($S_i \rightarrow 0$) marginal likelihood — but the S_i term is present, not dropped.

(B) Dropped from the E-step that builds $q(x)$ — the real omissions

The M-step conditions on the $q(x)$ produced by the SING $q(x)$ -update, and that update currently uses two approximations. They change (T, z_r) , hence $\mathcal{L}_M$ *indirectly*, but they are not terms of (7).

(B1) The drift-posterior-variance feedback $\mathbb{E}_{q(x_i)}[V_r]$ — piece (iv), “dropped (production)”. The exact transition contribution to the $q(x)$ -objective is (E-step doc eq. (transition-frozen))

$$\mathbb{E}_{q(x_i, x_{i+1})}[\log p(x_{i+1} \mid x_i, f)] = -\frac{\mathbb{E}[\delta_i^\top \Sigma^{-1} \delta_i]}{2\Delta_i} + \mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \bar{f}] - \frac{\Delta_i}{2} \mathbb{E}_{q(x_i)}[\underbrace{\bar{f}^\top \Sigma^{-1} \bar{f}}_{\text{(iii)}} + \underbrace{V}_{\text{(iv)}}]. \quad (13)$$

Production (`restore_qf_variance='none'`, “Approximation A”) **drops piece (iv)** $-\frac{\Delta_i}{2}\mathbb{E}_{q(x_i)}[V(x_i)]$ and, crucially, its (m_i, S_i) -gradient, from the natural-gradient $q(x)$ update. Concretely the omitted gradient is, by Price’s theorem,

$$\partial_{m_i}\mathbb{E}_{q(x_i)}[V(x_i)] = \mathbb{E}_{q(x_i)}[\nabla V(x_i)], \quad \partial_{S_i}\mathbb{E}_{q(x_i)}[V(x_i)] = \frac{1}{2}\mathbb{E}_{q(x_i)}[\nabla^2 V(x_i)]. \quad (14)$$

Meaning: setting $\mathbb{E}_{q(x_i)}[V] \equiv 0$ in the E-step makes $q(x)$ behave as if the drift were *known* (no epistemic-drift term inflating the posterior). The path is then reconstructed as tightly as the emissions allow. This is the exact sense of “EFGP does fixed-data GP inference”: the fixed data is the drift-uncertainty-free posterior mean. (`restore_qf_variance='hutch'` restores $\mathbb{E}_{q(x_i)}[V]$ via a Hutchinson estimate of $\omega_r(\delta) = \sum_{\xi_\ell - \xi_k = \delta} (A_r^{-1})_{k\ell} D_k D_\ell$; empirically here $\|\nabla \mathbb{E}_{q(x_i)}[V]\| \approx 5 \times 10^{-5} \|\text{legacy}\|$, so restoring it is inert — *this* problem is drift-well-determined.)

(B2) The Gauss–Newton / covariance-drop linearisation of the drift moments. For piece (iii), $g = \bar{f}^\top \Sigma^{-1} \bar{f}$, Price’s ∂_{S_i} needs $\frac{1}{2}\mathbb{E}_{q(x_i)}[\nabla^2 g]$; the Hessian splits (E-step doc eq. (gn-hessian))

$$\nabla^2(\bar{f}^\top \Sigma^{-1} \bar{f}) = 2(J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}} + \sum_r \sigma_r^{-2} \bar{f}_r H_{\bar{f}_r}), \quad (15)$$

and production makes two truncations, each second-order in the nonlinearity of $\bar{f}$:

$$\partial_{S_i}\mathbb{E}_{q(x_i)}[\bar{f}^\top \Sigma^{-1} \bar{f}] \stackrel{\text{GN}}{\approx} \mathbb{E}_{q(x_i)}[J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}}] \stackrel{\text{cov-drop}}{\approx} \mathbb{E}_{q(x_i)}[J_{\bar{f}}]^\top \Sigma^{-1} \mathbb{E}_{q(x_i)}[J_{\bar{f}}] \quad \left(\text{drops } \sum_r \sigma_r^{-2} \bar{f}_r H_{\bar{f}_r}, \text{ Cov}(J) \right). \quad (16)$$

Equivalently $\bar{f}$ is replaced by its first-order Taylor $\tilde{f}(x) = \bar{f}(m_i) + J_{\bar{f}}(m_i)(x - m_i)$ inside the $q(x_i)$ -average. Bias is $O(\|S_i\| \sigma_{\bar{f}}^2 / \ell^2)$: again small for $q(x)$ localised at scale ℓ .

5 Are the dropped terms easy or hard to compute?

Each dropped term, written explicitly, with its cost. *Nothing is dropped for lack of a closed form*; the reasons are “ θ -constant”, “already present”, or “cheap but chosen out for stability/variance”. The single genuinely non-trivial object is the global inverse A_r^{-1} inside the V -feedback *gradient*.

term	explicit form	cost / difficulty
(A) self-energy	$\frac{1}{2} \sum_i \Delta_i^{-1} \text{tr}(\Sigma^{-1} \mathbb{E}_{q(x_i, x_{i+1})}[\delta_i \delta_i^\top]),$ $\mathbb{E}_{q(x_i, x_{i+1})}[\delta_i \delta_i^\top] = S_{i+1} + S_i - S_{i,i+1} - S_{i,i+1}^\top$	trivial ($O(TD^2)$ traces of $q(x)$ moments) $d_i d_i^\top$ but θ -constant, so no reason to add.
(B1) $q(f)$ -var value	$\sum_i \Delta_i \mathbb{E}_{q(x_i)}[V_i] = D_{\text{out}}(M - \text{tr } A_\theta^{-1})$	free: $\text{tr } A_\theta^{-1} = \ L^{-1}\ _F^2$ reuses the M-step Cholesky $A_\theta = LL^*$; already implicit in $\log A_\theta $.
(B1) $q(f)$ -var feedback	$\partial_{m_i} \mathbb{E}_{q(x_i)}[V] = \mathbb{E}_{q(x_i)}[\nabla V],$ $\partial_{S_i} \mathbb{E}_{q(x_i)}[V] = \frac{1}{2} \mathbb{E}_{q(x_i)}[\nabla^2 V];$ needs $\omega_r(\delta) = \sum_{\xi_\ell - \xi_k = \delta} (A_r^{-1})_{k\ell} D_{\theta,k} D_{\theta,\ell}$	moderate–hard: A_r^{-1} is <i>non-local</i> (couples all M frequencies). Estimate ω_r by Hutchinson (P probes): $P \cdot D_{\text{out}}$ CG solves + per-source enveloped NUFFT. $\sim 5\text{--}15\%$ wall, noisy-but-unbiased. (= ‘hutch’.)
(B2) residual Hessian	$\frac{1}{2} \sum_r \sigma_r^{-2} \mathbb{E}_{q(x_i)}[\bar{f}_r H_{\bar{f}_r}],$ $H_{\bar{f}_r}(x) = (2\pi)^2 \sum_k \mu_{r,k} D_{\theta,k} \xi_k \xi_k^\top \phi_k(x)$	moderate: one $\xi_k \xi_k^\top$ -weighted Type-2 NUFFT (drift-moment machinery). Dropped for being <i>sign-indefinite</i> (breaks PSD natural-grad precision), not for cost.
(B2) $\text{Cov}_{q(x_i)}(J_{\bar{f}})$	$\mathbb{E}_{q(x_i)}[J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}}]$ $\mathbb{E}_{q(x_i)}[J_{\bar{f}}]^\top \Sigma^{-1} \mathbb{E}_{q(x_i)}[J_{\bar{f}}]$	easy–moderate: same $\xi \xi^\top$ -weighted NUFFT; $O(\ S_i\ ^2)$, dropped for simplicity.

Takeaway. The M-step objective’s own drop (A) is a θ -constant. Of the E-step drops: the V -value is free and already in $\log |A_\theta|$; the V -gradient (B1) is the only hard piece — hard purely because A_r^{-1} couples all M frequencies, so its per-source spatial imprint needs the FFT-cross-correlation+NUFFT trick rather than a local formula — and the GN residual/covariance terms (B2) are cheap and were dropped for PSD-stability and $O(S)$ -smallness, not intractability.

6 Consequence

(7) is the *exact* collapsed marginal likelihood of the drift regression for the Fourier model — it drops only the θ -constant self-energy, and it *keeps* the input uncertainty (12), the Stein correction, and the $q(f)$ variance (10). Therefore the length-scale behaviour of EFGP is *not* explained by a missing M-step term. What EFGP omits are the two E-step feedback terms (B1) $\mathbb{E}_{q(x_i)}[V]$ and (B2) the residual-Hessian/covariance, i.e. the drift’s epistemic uncertainty and the second-order effect of S_i on the drift moments. Both feed the M-step only through the $q(x)$ it conditions on, and both are $O(S)$ -or-smaller here. The net effect is that EFGP performs GP marginal-likelihood learning on a drift-uncertainty-free reconstruction — exact GP hyperparameter inference on “fixed data.”